

Artificial Intelligence 2

Exercise Sheet 3

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Individual Exercises (5 P)

- Visit <https://checkr.tugraz.at/> (a TU Graz software, login using your TU Graz account).
- At <https://checkr.tugraz.at/> enter the key 8ydg5H (same as for first exercise sheet).
- Provide your answers to the posed questions - several tries are possible (20 mins waiting time).
- If all questions have been answered correctly, this corresponds to 5 points (each correctly answered question corresponds to $\frac{1}{3}$ point).
- No explicit “Abgabe” is needed for this part (just answer the questions in KnowledgeCheckR).

Submission and Files

In the exercises below you will be asked to submit files in TeachCenter. Submit the files separately, not in archived (“zipped”) form. These are the files to submit:

- exercise3.pdf
- domain.txt
- problem.txt

Explain in your own words your solutions in exercise3.pdf.

Exercise 1: planning in PDDL

We are modeling a task of a robot in the PDDL language. The task we are modeling is for a robot who is a “support robot”, or SuBot for short.

SuBot helps people with their household and performs other duties. In our context, SuBot needs to serve dinner, fetch medicine, chat with their owner, and do some laundry. We utilize two constraints: *time* and *battery level*.

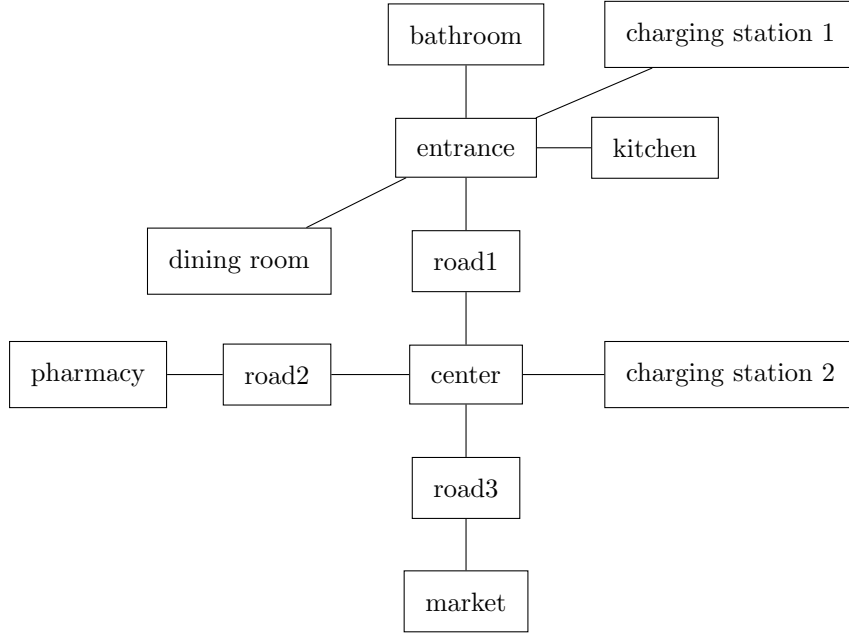


Figure 1: Reachability of locations

Locations and connections There are twelve locations in our context: a *bathroom*, a *kitchen*, a *dining room*, an *entrance hall*, three *roads*, two *charging stations*, a *pharmacy*, the “*center*”, and a *market*. These locations are connected as in Figure 1. For instance, we can move from the entrance hall to road1 and subsequently to the center. RoSP starts initially at the entrance hall.

Items There are three items of interest: medicine, groceries, and dinner. The medicine and the groceries are ready to be obtained from the pharmacy and market, respectively, and dinner needs to be prepared at the kitchen, provided that the groceries are carried.

Goal Our aim is to (i) serve dinner, (ii) place medicine in the bathroom, (iii) do laundry, and (iv) chat with the owner.

Time and Battery Level We model both time and battery level as (constant) functions in first-order logic. That is, (in the domain file) you need to specify time and battery-level functions. These will be increased, decreased, and possibly assigned as effects of actions, and can occur as preconditions in actions. We model both in a simplistic form: as natural numbers (including zero). Time starts at 1 and battery-level at 2.

Domain Consider the following information regarding the domain.

- Locations - places to move to (and perform other actions).
- Items - these can be obtained from the locations.

Actions Consider the following actions.

- Only you as the robot may take actions.
- You can move from one location to another, given there is an edge connecting them (see Figure 1). Moreover, if SuBot moves from one location to another, the battery-level is decremented by one and time is incremented by one. SuBot can only move if the battery-level is at least one.
- You can recharge at one of the two charging stations. This sets your battery level to twenty and increases time once.
- You can obtain the groceries from the market. Then you carry the groceries.
- You can fetch the medicine from the pharmacy. Then you carry the medicine.
- You can place the medicine in the bathroom, provided you are carrying the medicine.
- You can prepare dinner in the kitchen, provided you are carrying the groceries. Then you carry the dinner. Moreover, this advances time once.
- You can serve dinner. If you carry dinner then you can serve dinner in the dining room. This action needs to be taken within time slots 7 and 18.
- You can chat with the owner in the dining room, provided the dinner was served. This action must be taken within time slots 7 and 18. This action advances time once.
- You can do the laundry in the bathroom.

Modelling in PDDL

(7 P)

Model the previous description as a planning problem using PDDL:

- define the necessary atoms, constants, and predicates
- define all necessary actions including preconditions and effects
- describe the initial state and the goal
- write your description into a PDDL domain and problem file.

In your pdf file submitted to TeachCenter include an understandable documentation of your solution and, moreover, PDDL Domain and PDDL Problem files (the PDDL files should be separate files). These files should be executable in the online planning editor

`http://editor.planning.domains/`

Please note that this online editor is external. In particular, plan ahead in case there are unexpected downtimes.

Solver and requirements Please use “ENSHP: Expressive Numeric Heuristic Search Planner” at `editor.planning.domains`, or a solver which supports numeric values. You likely need to enter “(:requirements :strips :typing :negative-preconditions :numeric-fluents)” in the domain pddl file.

Solve and provide a plan (2 P)

Solve the planning problem using the online planning editor (see above) and report the obtained plan. This plan does not need to be optimal.

Would you use such a robot? (1 P)

Write in a few sentences in your own words what you think about a robot that supports people in their households. In particular, would you see it positively that issues of loneliness are addressed via robots?

Exercise 2: Bayesian Networks

In this exercise we are considering some aspects of Bayesian Networks.

D-Separation (2 P)

Consider the two directed acyclic graphs (DAGs) in Figure 2. Calculate for both DAGs, using the graph structures and d -separation only, the conditional independencies $I(X, Z, Y)$ (X is conditionally independent of Y given Z). To reduce the combinations, consider only $I(X, Z, Y)$ with

- X and Y being only single variables,
- Z can be a single variable or empty, and
- all three X , Y , and Z must be distinct.

Moreover, you can omit symmetric conditional independencies (e.g., $I(X, Z, Y)$ and $I(Y, Z, X)$ can be listed once) and those conditional independencies with Z not occurring between any path of X to Y (i.e., redundant Z 's can be omitted).

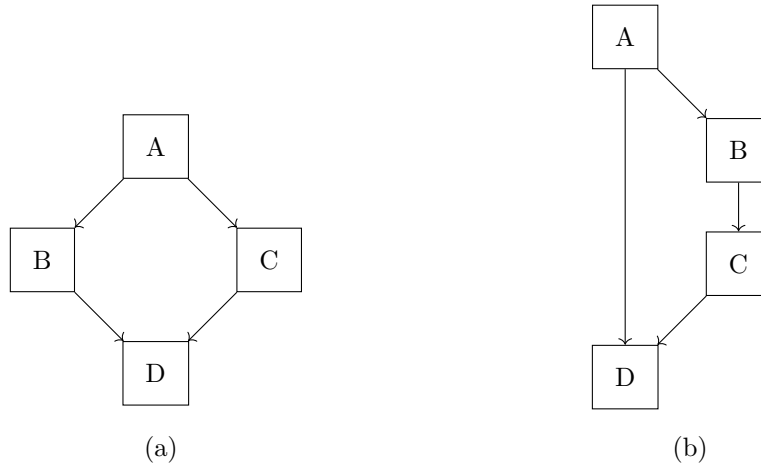


Figure 2: Two DAGs

Marginal Probabilities (0.5 P)

Compute, based on Table 1 the marginal probabilities of a positive occurrence of each variable (e.g., that A occurs positively).

Conditional Probability Tables (2 P)

Consider the joint probability distribution in Table 1. First, which of the two DAGs in Figure 2 represent this joint probability distribution? For the DAG that does not represent the joint probability distribution, give a justification why.

For your choice of DAG that does represent the joint probability distribution, compute the conditional probability tables of this DAG, given Table 1.

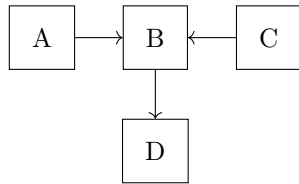
More on D-Separation (0.5 P)

Consider the graphs in Figure 3. Answer the following two questions.

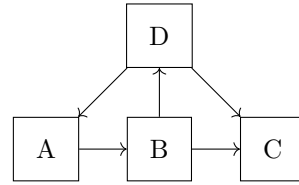
- For the DAG on the left, Figure 3(a), can you modify the graph by removing a single edge such that $I(\{A\}, \{D\}, \{C\})$ holds?
- Consider the graph in Figure 3(b). Is this a DAG?

Table 1: Joint probability distribution.

world	A	B	C	D	$P(\cdot)$
w_1	true	true	true	true	0.0192
w_2	true	true	true	false	0.0288
w_3	true	true	false	true	0.0864
w_4	true	true	false	false	0.3456
w_5	true	false	true	true	0.024
w_6	true	false	true	false	0.036
w_7	true	false	false	true	0.012
w_8	true	false	false	false	0.048
w_9	false	true	true	true	0.008
w_{10}	false	true	true	false	0.008
w_{11}	false	true	false	true	0.0432
w_{12}	false	true	false	false	0.1008
w_{13}	false	false	true	true	0.06
w_{14}	false	false	true	false	0.06
w_{15}	false	false	false	true	0.036
w_{16}	false	false	false	false	0.084



(a)



(b)

Figure 3: Two Graphs